```
# Install required libraries
!pip install transformers
!pip install torch
!pip install pandas
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/pyt
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Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/pyt
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Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/d
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.1
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-pack
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Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
```

```
# Import libraries
from transformers import pipeline, AutoModelForSeq2SeqLM, AutoTokenizer
```

```
import pandas as pd
```

```
# English to French
model_name_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_fr = AutoTokenizer.from_pretrained(model_name_fr)
model_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_fr)

# English to German
model_name_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

# English to Hindi
model_name_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_hi = AutoTokenizer.from_pretrained(model_name_hi)
model_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_hi)
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_
  warnings.warn("Recommended: pip install sacremoses.")
Loading weights: 100%  258/258 [00:00<00:00, 246.83it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared
The tied weights mapping and config for this model specifies to tie model.shared
config.json:        1.33k/? [00:00<00:00, 14.5kB/s]
tokenizer_config.json: 100%                    42.0/42.0 [00:00<00:00, 831B/s]
source.spm: 100%                               768k/768k [00:00<00:00, 13.1MB/s]
target.spm: 100%                               797k/797k [00:00<00:00, 43.4MB/s]
vocab.json:         1.27M/? [00:00<00:00, 28.3MB/s]
pytorch_model.bin: 100%                        298M/298M [00:03<00:00, 140MB/s]
Loading weights: 100%  258/258 [00:00<00:00, 573.00it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                        298M/298M [00:04<00:00, 63.1MB/s]
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The tied weights mapping and config for this model specifies to tie model.shared
generation_config.json: 100%                   293/293 [00:00<00:00, 17.3kB/s]
config.json:        1.39k/? [00:00<00:00, 72.6kB/s]
tokenizer_config.json: 100%                    44.0/44.0 [00:00<00:00, 2.75kB/s]
source.spm: 100%                               812k/812k [00:00<00:00, 2.40MB/s]
target.spm: 100%                               1.07M/1.07M [00:00<00:00, 3.07MB/s]
vocab.json:         2.10M/? [00:00<00:00, 18.7MB/s]
pytorch_model.bin: 100%                        306M/306M [00:02<00:00, 259MB/s]
Loading weights: 100%  258/258 [00:00<00:00, 641.86it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                        306M/306M [00:02<00:00, 205MB/s]
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The tied weights mapping and config for this model specifies to tie model.shared
generation_config.json: 100%                   293/293 [00:00<00:00, 25.7kB/s]
```

```
sentences = [
"Good morning, how are you today?",
"I will meet you at the library after class.",
"Where did you learn to speak French so well?",
"Can artificial intelligence improve healthcare systems?",
"The government announced a new policy to support farmers.",
"Heavy rainfall caused flooding in several coastal areas yesterday.",
```

```
    "Machine learning is a subset of artificial intelligence.",
    "Researchers are studying climate change and its impact on agriculture.",
    "The software uses a neural network to classify images accurately.",
    "Data encryption protects sensitive information.",
    "I just finished my assignment and feel relaxed.",
    "This smartphone camera is amazing.",
    "Although the weather was hot, students completed the project.",
    "She prepared the presentation carefully and the professor appreciated it.",
    "If technology evolves rapidly, communication systems may become automated."
    ]
```

```
for sentence in sentences:
    # French Translation
    inputs_fr = tokenizer_fr(sentence, return_tensors="pt", padding=True,
    translated_tokens_fr = model_fr.generate(**inputs_fr)
    french = tokenizer_fr.decode(translated_tokens_fr[0], skip_special_tok

    # German Translation
    inputs_de = tokenizer_de(sentence, return_tensors="pt", padding=True,
    translated_tokens_de = model_de.generate(**inputs_de)
    german = tokenizer_de.decode(translated_tokens_de[0], skip_special_tok

    # Hindi Translation
    inputs_hi = tokenizer_hi(sentence, return_tensors="pt", padding=True,
    translated_tokens_hi = model_hi.generate(**inputs_hi)
    hindi = tokenizer_hi.decode(translated_tokens_hi[0], skip_special_toke

    print("English:", sentence)
    print("French:", french)
    print("German:", german)
    print("Hindi:", hindi)
    print("----------------------------")
```

```
English: Good morning, how are you today?
French: Bonjour, comment allez-vous aujourd'hui ?
German: Guten Morgen, wie geht's dir heute?
Hindi: सुप्रभात, आप आज कैसे कर रहे हैं?
----------------------------
English: I will meet you at the library after class.
French: Je vous rejoins à la bibliothèque après le cours.
German: Wir treffen uns nach dem Unterricht in der Bibliothek.
Hindi: मैं क्लास के बाद पुस्तकालय में आप मिलेंगे.
----------------------------
English: Where did you learn to speak French so well?
French: Où avez-vous appris à parler si bien français ?
German: Wo haben Sie gelernt, Französisch so gut zu sprechen?
Hindi: आपने फ्रांसीसी भाषा इतनी अच्छी तरह बोलना कहाँ सीखा?
----------------------------
English: Can artificial intelligence improve healthcare systems?
French: L'intelligence artificielle peut-elle améliorer les systèmes de santé?
German: Kann künstliche Intelligenz die Gesundheitssysteme verbessern?
Hindi: क्या नक़ली बुद्धि स्वास्थ्यीय व्यवस्थाओं को बेहतर कर सकती है?
```

```
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English: The government announced a new policy to support farmers.
French: Le gouvernement a annoncé une nouvelle politique d'aide aux agriculteu
German: Die Regierung kündigte eine neue Politik zur Unterstützung der Landwir
Hindi: सरकार ने किसानों का समर्थन करने के लिए एक नई नीति घोषित की.
----------------------------
English: Heavy rainfall caused flooding in several coastal areas yesterday.
French: Des pluies abondantes ont provoqué des inondations dans plusieurs zone
German: Schwere Regenfälle verursachten gestern Überschwemmungen in mehreren K
Hindi: भारी वर्षा की वजह से कल के कई तटीय इलाकों में बाढ़ आ गयी ।
----------------------------
English: Machine learning is a subset of artificial intelligence.
French: L'apprentissage automatique est un sous-ensemble de l'intelligence art
German: Maschinelles Lernen ist eine Teilmenge künstlicher Intelligenz.
Hindi: मशीन सीखने के लिए कृत्रिम बुद्धि की एक नक़ल है।
----------------------------
English: Researchers are studying climate change and its impact on agriculture
French: Les chercheurs étudient les changements climatiques et leur impact sur
German: Forscher untersuchen den Klimawandel und seine Auswirkungen auf die La
Hindi: खोजकर्ता जलवायु परिवर्तन का अध्ययन कर रहे हैं और कृषि पर इसका प्रभाव डाल रहे हैं ।
----------------------------
English: The software uses a neural network to classify images accurately.
French: Le logiciel utilise un réseau neuronal pour classer les images avec pr
German: Die Software verwendet ein neuronales Netzwerk, um Bilder genau zu kla
Hindi: सॉफ्टवेयर छवियों को सही तरीके से बनाने के लिए तंत्रिका नेटवर्क का उपयोग करता है.
----------------------------
English: Data encryption protects sensitive information.
French: Le cryptage des données protège les informations sensibles.
German: Datenverschlüsselung schützt sensible Informationen.
Hindi: डाटा एनक्रिप्शन संवेदनशील जानकारी संरक्षित करता है.
----------------------------
English: I just finished my assignment and feel relaxed.
French: Je viens de terminer ma mission et je me sens détendu.
German: Ich habe gerade meine Aufgabe beendet und fühle mich entspannt.
Hindi: मैं अभी - अभी अपनी नियुक्ति खत्म कर चुका हूँ और तनाव महसूस कर रहा हूँ ।
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English: This smartphone camera is amazing.
French: Cette caméra pour smartphone est incroyable.
```

Start coding or generate with AI.

Start coding or generate with AI.